and describe how it would look from an unusual perspective such as the wheel of the bus, the pigeon in flight, or the quarter falling earthward. The challenge was to take an unfamiliar description like “I’m walking on a dark, cylindrical surface surrounded by low, textured walls, and an unruly bunch of thick white tendrils is descending from the sky,” and figure out that it was the view of an ant walking on a hot dog that a street vendor was garnishing with sauerkraut. Although we stopped playing years before I took my first physics course, the game is at least partly to blame for my having a fair amount of distress when I encountered Newton’s laws.

The game encouraged seeing the world from different vantage points and emphasized that each was as valid as any other. But according to Newton, while you are certainly free to contemplate the world from any perspective you choose, the different vantage points are by no means on an equal footing. From the viewpoint of an ant on an ice skater’s boot, it is the ice and the arena that are spinning; from the viewpoint of a spectator in the stands, it is the ice skater that is spinning. The two vantage points seem to be equally valid, they seem to be on an equal footing, they seem to stand in the symmetric relationship of each spinning with respect to the other. Yet, according to Newton, one of these perspectives is more right than the other since if it *really* is the ice skater that is spinning, his or her arms will splay outward, whereas if it *really* is the arena that is spinning, his or her arms will not. Accepting Newton’s absolute space meant accepting an absolute conception of acceleration, and, in particular, accepting an absolute answer regarding who or what is really spinning. I struggled to understand how this could possibly be true. Every source I consulted—textbooks and teachers alike—agreed that only relative motion had relevance when considering constant velocity motion, so why in the world, I endlessly puzzled, would accelerated motion be so different? Why wouldn’t *relative* acceleration, like relative velocity, be the only thing that’s relevant when considering motion at velocity that isn’t constant? The existence of absolute space decreed otherwise, but to me this seemed thoroughly peculiar.

Much later I learned that over the last few hundred years many physicists and philosophers—sometimes loudly, sometimes quietly—had struggled with the very same issue. Although Newton’s bucket seemed to show definitively that absolute space is what selects one perspective over